

Disk Scheduling Algorithms

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#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define N 8

int req[] = {98, 183, 37, 122, 14, 124, 65, 67};
int head = 53;

void sort(int *a, int n) {
    for (int i=0; i<n-1; i++) for (int j=0; j<n-1-i; j++)
        if (a[j]>a[j+1]) { int t=a[j]; a[j]=a[j+1]; a[j+1]=t; }
}

int abs_val(int x) { return x < 0 ? -x : x; }

/* — A) FCFS — */
void fcfs() {
    int total = 0, cur = head;
    printf("\n=== A) FCFS ===\nOrder: %d", cur);
    for (int i = 0; i < N; i++) {
        total += abs_val(req[i] - cur);
        cur = req[i];
        printf(" -> %d", cur);
    }
    printf("\nTotal seek: %d\n", total);
}

/* — B) SCAN — */
void scan() {
    int q[N]; memcpy(q, req, sizeof(req)); sort(q, N);
    int total = 0, cur = head;
    printf("\n=== B) SCAN (toward higher) ===\nOrder: %d", cur);
    // go right first
    for (int i = 0; i < N; i++)
        if (q[i] >= cur) { total += abs_val(q[i]-cur); cur=q[i]; printf(" -> %d", cur); }
    // go left
    for (int i = N-1; i >= 0; i--)
        if (q[i] < head) { total += abs_val(q[i]-cur); cur=q[i]; printf(" -> %d", cur); }
    printf("\nTotal seek: %d\n", total);
}
```

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/* — C) SSTF — */
void sstf() {
    int q[N], visited[N]; memcpy(q, req, sizeof(req)); memset(visited, 0, sizeof(visited));
    int total = 0, cur = head;
    printf("\n=== C) SSTF ===\nOrder: %d", cur);
    for (int i = 0; i < N; i++) {
        int min = 99999, idx = -1;
        for (int j = 0; j < N; j++)
            if (!visited[j] && abs_val(q[j]-cur) < min) { min = abs_val(q[j]-cur); idx = j; }
        visited[idx] = 1; total += min; cur = q[idx];
        printf(" -> %d", cur);
    }
    printf("\nTotal seek: %d\n", total);
}

/* — D) C-LOOK — */
void clook() {
    int q[N]; memcpy(q, req, sizeof(req)); sort(q, N);
    int total = 0, cur = head;
    printf("\n=== D) C-LOOK ===\nOrder: %d", cur);
    // serve all >= head
    for (int i = 0; i < N; i++)
        if (q[i] >= cur) { total += abs_val(q[i]-cur); cur=q[i]; printf(" -> %d", cur); }
    // jump to smallest and continue
    for (int i = 0; i < N; i++)
        if (q[i] < head) { total += abs_val(q[i]-cur); cur=q[i]; printf(" -> %d", cur); }
    printf("\nTotal seek: %d\n", total);
}

int main() {
    printf("1BM24CS361, Nitya R Swadi\n\n");
    printf("Disk head: %d\nRequests: ", head);
    for (int i = 0; i < N; i++) printf("%d ", req[i]);
    fcfs(); scan(); sstf(); clook();
}

```

OUTPUT:

```
"C:\Users\Nitya\OneDrive\De  X + v
1BM24CS361, Nitya R Swadi

Disk head: 53
Requests: 98 183 37 122 14 124 65 67
=== A) FCFS ===
Order: 53 -> 98 -> 183 -> 37 -> 122 -> 14 -> 124 -> 65 -> 67
Total seek: 640

=== B) SCAN (toward higher) ===
Order: 53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 37 -> 14
Total seek: 299

=== C) SSTF ===
Order: 53 -> 65 -> 67 -> 37 -> 14 -> 98 -> 122 -> 124 -> 183
Total seek: 236

=== D) C-LOOK ===
Order: 53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 14 -> 37
Total seek: 322

Process returned 0 (0x0)    execution time : 0.050 s
Press any key to continue.
|
```